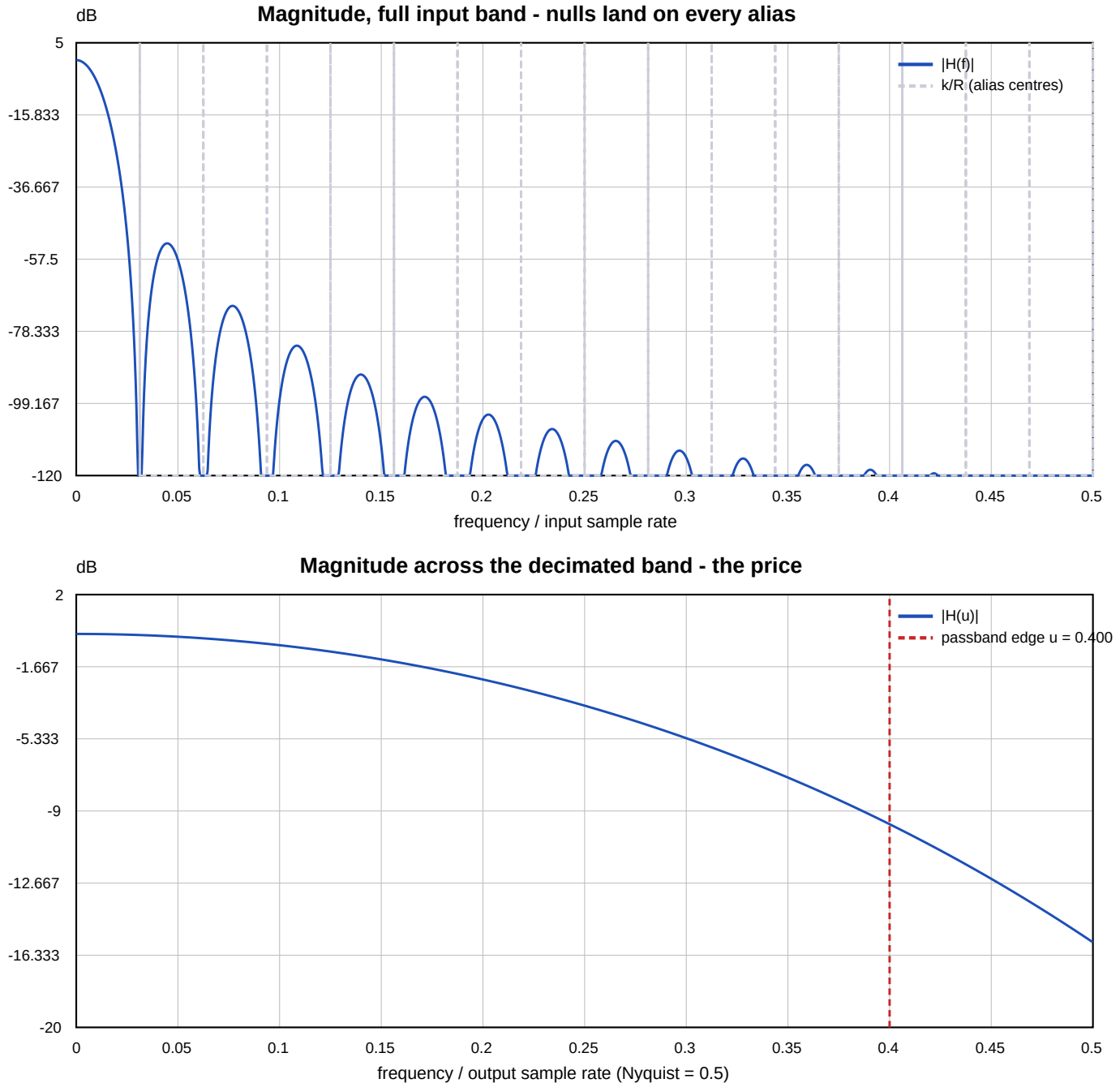


CIC Decimator - Frequency Response

N = 4 stages, R = 32, M = 1, input width 16 bits



Figures

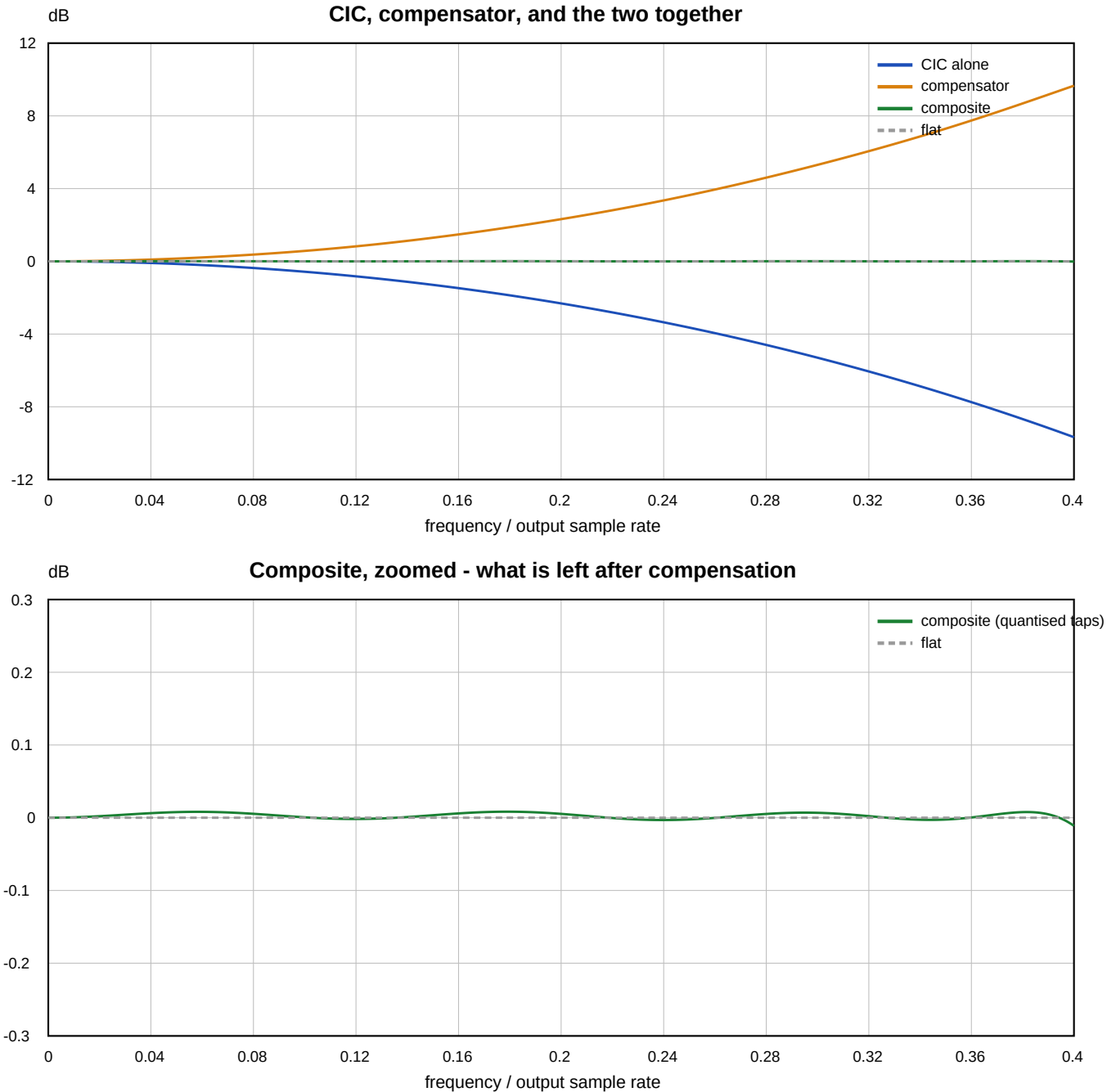
DC gain $(R \cdot M)^N$ 1048576
Accumulator width required 36 bits
Passband droop at $u = 0.400$ -9.67 dB
Worst alias in the passband -23.7 dB
First null at $f = 1/(R \cdot M)$ 0.03125

Pruned stage widths ($b_{\text{out}} = 20$) .. [35, 30, 26, 23, 21, 20, 19, 18]
Register bits: 192 pruned vs 288 uniform

Droop is inherent to the sinc^N shape, not a design error: the nulls that reject the aliases and the droop across the passband are the same expression. See page 2.

CIC Compensation Filter

15-tap symmetric FIR at the output rate, 16-bit coefficients



Figures

Uncompensated droop, DC to edge ... -9.67 dB

Ripple, ideal taps 0.0188 dB

Ripple, 16-bit taps 0.0195 dB

Improvement 497x

Peak gain asked of the filter 3.04x

Coefficient fractional bits 14

DC gain of quantised filter 1.000000

Multipliers (folded, symmetric) .. 8

Taps: [-37, 138, -373, 887, -1998, 4545, -11226, 32512, -11226, 4545, -1998, 887, -373, 138, -37]

Compensation shapes the passband only. Alias rejection is unchanged - if the aliases are too large the answer is more CIC stages or a narrower passband, not more taps.